

Yingpeng Ma

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EDUCATION

University of Stuttgart

Master of Science (M.Sc.) in Information Technology

Stuttgart, Germany

Oct 2020 – Feb 2024

GPA: 2.7 (German grading system, modified Bavarian formula; 1.0 = best, 4.0 = pass)

Commented [YM1]: 德国GPA计分4.0相当于60分及格，60分一下统一记为5.0.这里改为5.0=failed是不是好看一点？这个2.7的GPA离4.0=pass太近了... 显得分格外低...

University of Electronic Science and Technology of China

Bachelor of Engineering (B.Eng.) in Communication Engineering

Chengdu, China

Sep 2015 – Jul 2019

GPA: 3.41/4.0 (80.4/100)

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AWARDS

- First Prize, National College Students' Innovation and Entrepreneurship Competition
- Best Individual Scholarship (1st place among 20 students), Exchange Program at National University of Singapore (NUS)

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RESEARCH EXPERIENCES

Analyzing Human vs. Neural Attention in Visual Question Answering (VQA)

Master's Thesis

Aug 2023 – Feb 2024

Mentor: Prof. Dr. Andreas Bulling; Supervisor: Yao Wang

- Conducted a systematic study on the relationship between **human visual attention** and **neural attention mechanisms** in **Visual Question Answering (VQA)**, aiming to evaluate **model interpretability and transparency** in multimodal learning systems.
- Studied foundational theories of **Graph Neural Networks (GNNs)**, **human attention modeling**, and **machine attention mechanisms**, establishing a solid theoretical basis for human-AI attention comparison.
- Explored **explainable AI (XAI)** techniques using **AWS**, **Google Colab**, and **Hugging Face Playground**, and reproduced and benchmarked **five state-of-the-art CQA/VQA models** to ensure experimental reliability.
- Extracted **machine attention maps from model output layers** and reconstructed them into **human-interpretable grayscale visualizations**, enabling direct qualitative and quantitative comparison with **human-annotated attention maps**.
- Designed an **end-to-end experimental pipeline** for attention extraction, visualization, and comparison, ensuring reproducibility and systematic analysis across different models.
- Performed correlation and **structural similarity analysis** between machine-generated and human attention maps, demonstrating a **strong alignment between neural and human attention patterns**, thereby providing empirical evidence for **model interpretability and human-computer alignment**.
- Implemented the full research workflow using **Python**, with **Git-based version control**, **Jupyter Notebook-based visualization**, and interactive experimentation to support efficient research iteration.

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Visualization for Computational EEG Methods

Graduate Seminar Project

Apr 2022 – Oct 2022

Mentor: Jun.-Prof. Dr. Benedikt Ehinger

- Studied and implemented core **computational EEG preprocessing and analysis methods**, including **Independent Component Analysis (ICA)**, **baseline correction**, and re-

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referencing, to support robust neural signal interpretation.

- Developed **interactive and reproducible visualization pipelines** to illustrate the effects of different EEG preprocessing steps, enhancing interpretability and methodological understanding.
- Designed **interactive computational notebooks** that allow dynamic parameter adjustment and real-time visualization, facilitating intuitive exploration of EEG signal transformations.
- Implemented the entire workflow using **Julia programming language**, leveraging **Julia Pluto Notebooks** for **interactive visualization and exploratory analysis** in a reproducible research setting.

Image Saliency Detection with Bit-Plane Slicing

Undergraduate Research Project

Sep 2017 – Jun 2019

Mentor: Prof. Dr. Bing Zeng

- Studied the **fundamental research methodologies, theoretical frameworks, and experimental workflows** in computer vision, with a focus on **image and video representation and analysis**.
- Learned and reproduced **classical image and video encoding techniques**, and analyzed both **traditional and machine learning-based saliency detection and image segmentation methods**.
- Systematically evaluated existing saliency detection algorithms in terms of **accuracy, interpretability, and computational/resource requirements**, highlighting trade-offs relevant to practical deployment.
- Proposed a **novel saliency detection algorithm based on image bit-plane (bitmap) slicing**, aiming to reduce computational complexity while preserving salient structural information.
- Designed and implemented a **lightweight machine learning-based saliency detection model**, optimized for **resource-limited environments**.
- Conducted comprehensive comparisons with representative **traditional algorithms and ML-based models**, evaluating **runtime, memory footprint, and Mean Absolute Error (MAE)**, and demonstrated **competitive performance under constrained computational budgets**.

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内容必要性不大，但是作为本科项目，较为基础，为体现基础科研能力和学习能力，可以合并为简短的一条

PROJECTS

GazeEval-VLM (Open-Source Project)

Independent Project (<https://github.com/ingingX/GazeEval-VLM>)

Independent Researcher for Human-Centered AI

Mar. 2026 – Present

- **Project Description**
- Focused on the development, integration, and benchmarking of a **model-agnostic, human-gaze-guided vision-language model evaluation framework**, analyzing the impact of **cognitive signals** on VLM performance across complex **visual and chart reasoning tasks**.
- **Work Description**
- Designed a **reproducible experiment pipeline** and **multimodal AI system** using **Python, PyTorch, and Hugging Face Transformers** to systematically process **human gaze data** and optimize **VLM inference pipelines**.
- Devised GazeVLM-Lite, a **lightweight inference methodology** introducing diverse **gaze-guided visual interventions**, including **gaze crop, gaze highlight, blurred background**, and

multi-view inputs, to provide robust **visual evidence** and significantly enhance model performance.

- Executed a comprehensive **cross-domain comparative study** across various **VQA** and **CQA** sub-tasks, quantitatively evaluating the differential impact of **gaze guidance** and task reliance on **local visual evidence** versus **global visual context**.
- Conceptualized a novel **gaze efficiency score metric** to mathematically evaluate the trade-off between model accuracy improvements and the additional **attention resource consumption** introduced by cognitive signals.
- **Constructed** data processing, result tracking, and **interactive demonstration interfaces** leveraging **OpenCV**, **NumPy**, **Pandas**, **Matplotlib**, and **Gradio** to support **Explainable AI (XAI)** and **visual attention modeling**.

LatentFit (Open-Source Project)

Independent Project (<https://github.com/ingingX/LatentFit>)

Independent Researcher

Sep. 2025 – Dec. 2025

- **Project Description**

- Directed the development of a lightweight **semantic image retrieval system** based on the **Fashion-MNIST dataset**, building an **embedding learning pipeline** to capture **embedding similarity** through robust **representation learning**.

- **Work Description**

- Orchestrated an end-to-end **research-oriented ML prototyping pipeline**, independently managing the complete project lifecycle from data curation to **embedding retrieval** and experimental documentation.
- Trained a **Triplet CNN Embedding Encoder** and optimized **Triplet Metric Learning** utilizing **PyTorch**, **torchvision**, and **GPU-based deep learning**, enabling the network to learn highly discriminative **semantic embeddings**.
- Analyzed complex **embedding geometry** and **semantic clustering** by establishing an **embedding extraction pipeline** and performing **t-SNE dimensionality reduction** with **NumPy**, **scikit-learn**, and **Matplotlib**.
- Realized a functional **Semantic Retrieval Demo** by leveraging the learned representations, demonstrating highly efficient and accurate **similarity search** capabilities within high-dimensional vector spaces.

ValveSense-v2 (Open-Source Project)

Independent Project (<https://github.com/ingingX/ValveSense-v2>)

Independent Researcher

Sep. 2024 – Nov. 2024

- **Project Description**

- Initiated an industrial **time-series forecasting framework** driven by **CEEMDAN** and **SVR**, enabling the non-linear prediction and analysis of real-world **industrial sensor data** through a fully integrated experimental evaluation and interactive deployment system.

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- **Work Description**

- Established a modular **system architecture**—encompassing Decomposition, Feature Engineering, Forecasting, Evaluation, Visualization, and Deployment layers—to deliver a **scalable machine learning framework** and robust **research-oriented software engineering**.
- Integrated **CEEMDAN signal decomposition** and **IMF-level forecasting** utilizing **Support Vector Regression (SVR)** with an **RBF kernel**, optimizing **non-stationary signal analysis**, **multi-resolution prediction**, and accurate **signal reconstruction**.
- Structured a rigorous **benchmark experiment system** for **recursive future forecasting**, independently overseeing the entire **AI pipeline engineering** process from data preprocessing to containerized deployment using **Docker**, **WSL2**, and **Git/GitHub** for **CI/CD** version control.
- Delivered an interactive **AI Web application** powered by **Python**, **Gradio**, and **Streamlit** (supported by **NumPy**, **Pandas**, **Matplotlib**, **scikit-learn**, and **PyEMD**), successfully demonstrating **AI product prototyping**, **interactive ML systems**, and **Human-in-the-loop AI** principles.

Sanhua Automotive Europe GmbH

Full-time Advanced System Engineer & IT Specialist

Stuttgart, Germany

Apr. 2024 – Present

- **Project Description**

- Focused on the **development, integration, and deployment of intelligent enterprise information systems**, supporting secure data exchange, conversational AI services, and **large-scale ERP system upgrades** in an industrial environment.

- **Work Description**

- Designed and implemented **secure and reliable APIs** to enable accurate data transfer between **Weaver OA systems** and **OpenAI-based AI services**, leveraging **JavaScript**, **SQL**, **FastAPI**, and **low-code development platforms** to ensure scalability and maintainability.
- Deployed and fine-tuned **task-oriented conversational AI models**, optimizing performance across diverse enterprise use cases using **Python** and **JavaScript**, with a focus on robustness, response quality, and system stability.
- Actively participated in the **INFOR LN ERP system upgrade**, developing **system integration modules**, **custom APIs**, and **tailored functional extensions** to support business-specific workflows and data pipelines.
- Provided **on-site and remote IT administration and troubleshooting**, covering **hardware and software systems**, ensuring operational continuity and efficient technical support across departments.

Sanhua Automotive Europe GmbH

Part-time Werkstudent – Junior IT & Digitization Support

Stuttgart, Germany

Jan. 2022 – Sep. 2023

- **Project Description**

- Provided comprehensive **internal IT and digitalization support** across departments, contributing to the stability, security, and continuous improvement of enterprise IT

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升级和申请项目关系不大，可以改为industrial automation script and system development. 但是和后边的 industrial environment有点重复。麻烦老师看着组织一下表达。

Commented [YM9]: 这两段可以换一下位置，凸显AI system的部署能力。

Commented [YM10]: ERP 这个删了，改为automation script and system development. ValveSense-v2 项目是这个system development的一部分。

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infrastructure.

- **Work Description**
- Delivered **multi-level IT support services**, resolving **hardware, software, and network-related issues** for end users in daily operations.
- Managed and maintained **Windows Active Directory (AD)** environments, including user account administration, access control, and policy configuration.
- Participated in **IT infrastructure upgrades**, supporting system updates, hardware replacements, and software deployments to improve reliability and performance.
- Assisted in **enterprise digitization initiatives**, contributing to the modernization of internal IT workflows and operational processes.

SKILLS

- **Deep Learning & Machine Learning:** Proficient in designing, training, and evaluating models using **PyTorch** and **TensorFlow**, with experience in **CNNs, RNNs, Transformers, GNNs**, and **diffusion-based models**.
- **AI Research & Modeling:** Strong background in **model architecture design**, **loss function engineering**, **optimization strategies**, and **experimental benchmarking** for research-oriented AI systems.
- **Programming & Systems:** Proficient in **Python, Java**, and **C**, with solid experience in **algorithm development**, **numerical computing**, and **system-level implementation**.
- **Data Processing & Experimentation:** Skilled in **data preprocessing**, **dataset construction**, **statistical analysis**, and **reproducible experimentation**, with extensive use of **NumPy, SciPy, Pandas**, and **Jupyter notebooks**.
- **Scientific Computing & Writing:** Experienced with **MATLAB, Mathematica**, and **LaTeX**, supporting both **engineering computation** and **academic research dissemination**.

INTERESTS

- **Electrical Engineering (Hands-on Practice):** Practical experience with basic electrical systems, circuit wiring, and troubleshooting, reinforcing engineering intuition and problem-solving abilities.